

Ideal Help Desk/Service Desk in E-Government and Service Quality: A Literature Review

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Abstract—Help desk and service desk are critical in this era regarding how much people need a place to ask to and consult to. People demand a transparent governance and help desk service desk is one way to fulfill that demand. But, there's ambiguity in what is the ideal help desk, because the condition in every country is different, they have their own characteristic. There are some countries that has made services for their citizens full on electronic-based, while some still retain the conventional help desk service. There are two kinds of e-government which we have known, developed countries and developing countries. And they have their own design of what help desk they should use, but some are not very effective and efficient as it was expected to be. This article discussed about what is the ideal help desk/service desk in e-government and the service quality as the indicator to know whether that help desk has been used on its maximum advantages or not.

Keywords—E-Government; Service Desk; Help Desk; Service Quality

I. INTRODUCTION

This article provides some discussions about what is the meaning of help desk and service desk and what its use and benefits for E-Government or Smart city. As we know, now in this era, E-Government is already being implemented in some nations, such as some advance nations like Japan, US, German, etc. Some reasons about why help desk and service desk is important are: (1) citizen's demand for transparent governance, (2) citizen's knowledge for some of government's rule and system is still limited, and (3) their needs for government's advice and fast response to things around them. Some additional reasons are provided about why do we need good help desk and service. Because some nations have their own ideal help desk, and every of them have their beneficial reason, E-Government can gather all the positive things in help desk service desk that has been implemented and decide what is the best beneficial and informative help desk they can make in the future. This article proposes the ideal help desk service desk based on other nations' experiences and successfulness in hope that it could be in help for the E-Government.

This paper consists of four sections. The next section is about the base of this review, which is E-Government, ITIL, and help desk or service desk. The third consist of discussion

about what kind of help desk that has been used and implemented in some advanced countries, why do we need help desk and service desk for E-Government, what are the advantages and disadvantages from the use of some kinds of service desk, and some criteria to determine service quality. From the information that has been gathered in section three, section four conclude that into a conclusion about what is the ideal help desk for E-Government and what kind of help desk that is best fit regarding the condition of the countries.

II. METHODOLOGY

A. E-Government

E-government can be defined simply as using ICT to improve the process of government [1]. To put it simply, it sometimes can be called as citizen's service, or re-engineering with technology [2].

From the definitions above, e-government is automation of computerization in governance and procedures that will drive a new kind of ways in transacting, business, listening to and answer to citizens, and organizing and delivering information [3]. E-government's goal is to enhance access of service to citizens' benefit. And also, to increased effectiveness in governance and transparency to become a better government.

In this way, citizens got high benefit from e-government, because one of e-government goals is to enhance service. The kind of service e-government offer should be effective and satisfying. There are some potential advantages citizens can get from e-government: round the clock government, no need to tell the government and filling form a number of times but only once is enough (one-stop shops), electronic advantages where there's no need to queuing in information sharing, and more open and cheaper government (better public access) [4].

There is no one fit and perfect model for e-government nor universal standard for e-government readiness because every nation has their own needs and scale of priorities [5]. Their citizens' most important needs influence e-government act. So, the services that e-governments offer are not the same in some nations. They offer the services that they think meet their people demand and give advantages for future, effective and efficient services.

B. ITIL

ITIL (Information Technology Infrastructure Library) is a set of service management standard library, focused on IT industry but can be implemented in organization and e-government. Developed by the CCTA (Central Computer and Telecommunications Agency) merged with the OGC, in the 1980s of the UK government [6].

The highlight of ITIL is on managing service from citizens or customer desires and the organization, here it's e-government. ITIL contain five service delivery processes, five service support processes, and one service support function (service desk) [7]. In this case, we highlight the help desk, or service support function.

The use of ITIL can boost e-government effectiveness in delivering service to citizens. ITIL served as a framework for service desk/help desk and the staff's duty. ITIL implementation in a nation will let help desk track and register the whole process, and assure practical resolution to citizens' problems [6].

ITIL best practice framework allow managers to record, review, and improve the service management processes [8]. This means a lot of benefits for e-government too, regarding the aim of e-government is to use IT to improve their service for citizens. ITIL helps e-government to meet its objectives. By recording the processes, it also simplifies the search of suitable answer or service and can be used in making Frequently Asked Question.

Many nations and organizations have adopted the ITIL in their management service to serve effective management and control in service support and delivery. ITIL can be used as a guideline of quality service for objectives: better quality control, service level, minimize cost, efficiency, and information flow effectiveness. The environment (social, organizational, and physical), processes, and interdependencies among other nations should be deliberated in making the ITIL framework that suit them [9].

C. Help Desk/Service Desk

Service desk is defined as a single point of contacts between the service provider and users. It also manages service request and handles communication with users [6]. Provider means the one who deliver the service, e-government act as a provider, while the citizens act as users.

Service desk and help desk is kind of the same things defined by different names. Help desk also defined as an area in any organizations which help end-users find answers and solution to problems they face [10].

Help desk is a specific type of service, an outsourced support group aiding callers by resolving the problem or transfer the call to other for help. Typical help desk workload consists of solution to the problem and containment of crisis. Most of them in functioning in the reactive area of support only [11].

As the function of help desk is to serve the citizens and resolve their problem, it is important for help desk to have

friendly attitude and knowledge of the problems that has been asked to them. Some elements that are the most important to users are technical knowledge, friendliness, quick service, and implicit communications [11]. Most users demand to know the answer as quickly as possible and clear communication will not lead them to endless confusion. Help desk should clarify users' doubt. So, it is important to have the competent staff to handle help desk.

III. STUDY

A. Help Desk for E-Government

Why do we need help desk in e-government? Help desk service, as cited before, is crucial for e-government, as e-government is built from services to give benefit for citizens, business partners, etc. we need help desk services in various kind to meet people demand, to build a transparent governance, and to make sure citizens meet what they need.

The strategic objective of e-governance is to support and simplify governance for government, citizens, and businesses. The purpose of the use of ICT is to connect all of them, while giving support in processes. The external goal of e-government is to fulfil people's need and their expectations of satisfaction on the front-office side by simplify the interaction with some kinds of online services [5]. By using ICT, the government serve speed, transparent, efficient, and effective interaction with public and other. The back-office's objective is to facilitate those serve. By doing that, the cost can be minimized.

E-Government engross the use of ICT to serve citizens with government information and services, most prominently through the around the clock availability of the World Wide Web [12]. Help desk in e-government is a one-way tool for people to ask something they don't know regarding government. Government is an organization with a high complexity. Their service for their people, such as how to make driving license, require some steps to get so not everyone understand that. If they want to ask something outside the working hour, they need some mediator to deliver their question or aspiration to the government.

But, as the demand for help desk/service desk arise exponentially, people need a good help desk that is really useful for them, with fast response and reliable. While e-government need help desk that is effective and efficient so that it hit the right spot with minimum amount of money for it.

There are two ways in which help desk are handled, centralized and distributed. Some studies recommend a centralized help desk. Centralizing help desk increased efficiency, give more consistent service, and economy of scale. But others recommend distributing service to individual departments for a better response and improve understanding level between users and technical staff [10]. So, they both have their advantages. But for efficiency, centralized help desk is better than distributed.

To implement help desk, we need to know help desk management, which are: training time of help desk staff,

citizens complain amount, delayed response to service request's percentage, the percentage of service request record correctly described, the percentage of correct record of help desk staff [6] to manage and keep our help desk on track.

There is also other way to handle help desk, by outsourcing it to vendor outside e-government, the example of this is application "Qlue" that has been used in Jakarta as an e-government help desk application. Decision makers of e-government might deliberate outsourcing any or all help desk functions, considering the resources in the so-called e-government [13].

The problem in outsourcing help desk service is that under performance and low competency skills will give service levels lower than what expected in service level agreement established between user and outsourcing vendor will create delay in resolve the customer's problem. Service levels are performance indicator of service performed by the service provider [6]. If this case happened, it will reduce the trust and comfort public expected from e-government.

B. Help Desk/Service Desk in Other Nations

Some other nations have various kinds of help desk. Because there's not a nation which has the same condition and needs as the other nation, the kind of help desk that they use as an ace is not really the same. But most of developed nations use various kinds of help desk for the sake of their people comfort and trust.

In developed countries, it's not a big matter to imagine interaction form citizens with government be done 24 hours and seven days without having the difficulties of queue for a service. But the condition is not the same in developing countries, this level of efficiency and effectiveness is a bit difficult to reach. The condition is not supporting that. But on some experience in developed countries, it is possible to do that if governments disposed to decentralized responsibilities and processes and use electronic means [5].

By using electronic means, each citizen can contact the government via website, in which all forms, legislation, news, FAQ, and other information available. Developing countries can try this trend that has been used in developed countries to serve its citizens and businesses better, and to minimized the costs by making internal operations more efficient, as a collector and source of information.

Some countries have developed service of full online communication and service handling, while others are only offering basic information about government. Geographical distance and condition may encourage people to use and access e-government service, in some countries, in other countries, they may divert citizens service from web to call center to service desk and already become successful in multichannel approach [4]. Some quality measurement such as user-friendliness may influence citizens to use e-government service.

In Dutch, their citizens only use simple e-government services, their main function of online public service delivery is providing information. In city level websites, information

like addresses and telephone number are mainly searched, while in national level, citizens mainly looking for FAQ [4].

This is some of help desks that has been used broadly in developed country:

1) Call Center

Call center can be defined as a central location that is primarily used to receive phone calls from customers [11].

It is the most used help desk in every nation. Because of its advantages where we can get answer fast and directly. A call center is handled by an operator or receiver. Because it is connected directly with the people in government and become the main help desk because of its simplicity, it is significant to have someone competent handled it and become the operator to interact with clients.

The process of call center is citizen with a technical problem contacts the help desk by calling a toll-free number [14]. The people mainly handled this is help desk staff, and they speak directly with citizens, so it's important to train the staff for the call center continuity. Their friendliness and clear answers will decide the continuity of this help desk.

Call centers should employ people that can easily adapted to different shifts and the cultural aspects of call center work [11]. With poor outsourcing services, the customers give up or seek help elsewhere.

A lot of nations use this kind of help desk, like USA, Singapore, Japan, Russia.

2) Chat

Can be handled by few operators, chat is one simple way to deliver our question as it is ordinary and doesn't need internet connection. Its simplicity and non-time consuming has made chat become a popular addition to the service available on help desk's home page.

A person doesn't need to speak when seeking help through chat, so it is more quiet and suitable for people who doesn't like to speak or are having different ability.

Chat-based customer service features are available on many marketers' sites and led the current central help desk customer to expect that option there as well [15]. The non-electric chat is text messaging. Not very well-known in developed countries as the use of ICT make it easier.

The example of this help desk is on USA's website, <https://www.usa.gov/chat>, where we can send messages through it.

3) Emails

Emails is one of the alternative help desk service media. As it is cheap and most people have them. It's served as a basic media communication through electronic means.

Emails can also be linked with other help desk as a register identity. And the memory capacity is pretty big. Emails also can be used to give formal answer to customer.

As one of the most common way to communicate, it is easy to find government email on their website or contacts site.

4) Information Service Center

Information service center, or help center, is a help desk with a wide range of responsibility, with the aims to provide faster service and improve the satisfaction of customer. The faster solutions are delivered, the satisfaction rise higher. By this, help center need to have the ability and the authority it needs to handle [16].

Citizens can have direct contact to seek answer by using this service. This help center is nearly the same as customer services in bank.

By having direct contact, citizens' trust toward government rise. Just like call center, it's important to have competent people to handle this because customers' satisfactory rely on them.

The weakness of this kind of help desk is, it's difficult to have an open office for service 24 hours per seven days. And citizens have to wait in line to get the service.

Hong Kong use this kind of help desk.

5) Website

Websites are assumed to be the primary method of Internet-based interactions and service [17]. Websites can hold a lot of information regarding the government, such as services, news and other information. FAQ can be found here too. Many nations use it as the main help desk services.

As an internet-based service delivery, it includes aspects of the underlying application, user interface design, navigation, information quality, and service quality. The importance of these aspects can change over time. Website user's satisfaction depends on website functionality, ex: the number and quality of functions and value-added services offered [17].

Some governments use official website for help desk. some of them are <https://www.usa.gov/>, <http://www.gouvernement.fr/>, <https://www.gov.sg/>, <http://www.japan.go.jp/>, those in chronological are these nations governments' (USA, France, Singapore, Japan) official website.

6) Application

Application is new kind of help desk service, not many governments use this because of it is difficult and pretty costly.

However, it is one of the easiest way of service delivery, because additional features can be added here. And a lot of people interact with their mobile phones more than with their laptops. So, this make service delivery simpler and easier.

Applications can also help manage service level agreements. The problem in application is if any software application or module in IT service behaves incorrectly, service quality, and customer satisfaction will decrease significantly [18].

7) Official Account (OA)

A lot of people use media social as a way to communicate with others. So, official account can become important. It also makes the interaction between citizens and government easier.

OA can also be used to spread news and official announcement. This OA can spread other information publicly too. And can be used to introduce other kind of help desk to citizens.

C. Advantages and Disadvantages

TABLE I. ADVANTAGES AND DISADVANTAGES

Type	Details	
	Advantages	Disadvantages
Call Center	Commonly used, easy to use for the public, do not need the internet connection, information can be get fast and easy.	Need a fair amount of money expense for the call center operator and information desk people, need to train the employees to be fast and reliable to manage call center and answer it.
Text Messaging	Common, easy to use, can do not need internet connection, can be used for people who feel insecure to talk via call center or people with diffability, simple.	User's identity can cause ambiguity, there could be inaccuracy in sending the message (human error), high risk of getting a lot of spam messages.
Information Center Office	Information center is highly trusted, critics and question can be gathered fast and transparent, the questioner identity is clear.	Need fair number of employees and workplace as an information center office, the service can't be done 24 hours per seven, and cannot be done for a lot of people concurrently because it's strict by the number of employees and place.
Website	Doesn't bound by distance and time by using internet connection, the service can be held 24 hours a day, it can be used for a long time, doesn't need a lot of worker to manage it, there's easy access to connect to website, especially for a city with a high bandwidth, additional information and FAQ (Frequently Asked Questions) can be added in it.	Website's user interface is the key point for a lot of people, less interesting interface and rarely updated website will reduce people's trust and satisfaction towards e-government services. The layout and interface's simplicity from a website play an important task for user's comprehension and the tendency to access that website again. People also need internet connection to access a website so it's not very suitable for a region which has slow connection.
Help Desk Application	Easy to get and use, easy to send questions with many choices. Additional features such as GPS and call can be added to make user comfortable and	The making of help desk application need a significant amount of time and money. Because an application is not something that can be

Type	Details	
	<i>Advantages</i>	<i>Disadvantages</i>
	simplify some places added. Questioner's identity can be verified by using e-mail or phone numbers. Some additional features can be added such as FAQ and other can be added too and can be accessed without internet.	implemented without testing and so on
Official Account	Easy to use and to ask, spread fast and have a broad area, can be used as a place to announce things, questions that has been asked before are displayed and can be seen by everyone who connect to it.	The security is the problem, high risk of hacking and easy to get spams. Accounts' post determined citizen's impressions, every good or bad messages regarding e-government can be seen by everyone.
E-mail	Used by a lot of people, the basic electronic communications, cheap and fast	High risk of spam, insecurity of getting hacked, not so flexible.

Just like what mentioned at Table 1, there are many kinds of help desk service with their advantages and disadvantages. They can be categorized in two, ones that need internet connection and ones that doesn't. Ones that need connection are Official Account, help desk application, website, and e-mail. The advantages of these is we can give fast responses and real time, easy and simple. The problem of this lies on the internet connection. There are regions in some countries where it's hard to get a good signal, moreover internet connections, so the use of these help desk might not be effective and people can't have its benefits fully.

The others which doesn't need connections are call center, text messaging, and information center office. This kind has advantages of generally known and used, so it can be used and placed nearly everywhere. The minus is that it's need personnel who manage that and become call center employee and because it is handled manually by human, there can be data inaccuracy in process and high risk of error, this kind is not so effective to handle a lot of question at once.

D. Service Quality

To find out whether the service is effective and good or not, there should be standard in quality. The quality of e-services approaches focuses on the quality of the service itself as delivered. The approach is customer oriented because the service is based on customers' need and demand. Quality dimensions of these are related to the delivered service (availability, usability, security etc. of the service) and input from the receivers of the service (customers' priorities and needs) [19].

Some criteria that can be used to determined e-government service quality are [19]:

- Ease of Use (navigation, personalization, technical efficiency)

- Trust (privacy, security)
- Functionality of the Interaction Environment (support in completing forms)
- Reliability (accessibility, availability)
- and Content and Appearance of Information and Citizen Support (Interactivity).

These criteria can be used to measure the quality of help desk. Let's take website for example, a website is considered a good website if it is reliable, easy to use, simple, have a good privacy, and fast response. These criteria can be applied on other help desk too.

E-government aims to give quality service to citizens, it's related to how citizens perceive e-government. Other study also said that the most consistent predictors of citizens' attitude to e-government are their perception of its relative advantage, compatibility (with the citizen's existing values, beliefs, experiences, and needs), usefulness, and ease of use [20].

Citizens also expect the availability of service 24 hours per 7 days. It can't be implemented on a kind of help desk like help center, but can be implemented in electronic based help desk.

IV. CONCLUSION

The ideal help desk is help desk that can fulfill citizens' need, reliable, and suited to e-government condition. E-government should consider what citizens want. Some elements that are the most important to users (citizens) are technical knowledge, friendliness, quick service, and implicit communications. Help desk should clarify users' doubt. So, it is important to have the competent staff to handle help desk.

The ideal help desk at least consists of website, call center and email. Because it's like the basic standard and help desk central that people used and access frequently.

Websites are assumed to be the primary method of Internet-based interactions and service. Websites can hold a lot of information regarding the government, such as services, news and other information. FAQ can be found here too. Many nations use it as the main help desk services. Call center is the most used help desk in every nation. Because of its advantages where we can get answer fast and directly. Email has big memory to store questions and not a synchronous communication, so it makes operator job easier. And email is the basic media communication that is cheap and are used frequently.

Other help desk just like application and so on is good to apply to e-government if the resources and citizens support it. If every help desk is implemented in a nation, means the citizens and resources of the nations is particularly good at responding and support government.

Other help desk service can be implemented based on the government's need and people's demand. Call center and

website can't serve a mass number of users, so the other help desk can be the alternative solution.

E-government should manage help desk/service desk on its own so the government won't get the risk of miscommunication between the workers and the answers to the question could become on point. E-government should consider and understand its own nation's condition to make effective and efficient help desk.

To make a good help desk, e-government can make one division to particularly manage help desk service. So, e-government can assign people for official accounts operator, information center employees, and call center receiver. With fast and satisfied work and response, e-government service will be acknowledged by their people and e-government efficiency and effectiveness level will rise. The opportunity for e-government work and service to be better will rise because people critics and suggestion, as well as question about things they don't know, are accepted and responded well by the government. The government transparency raises the people trust and people demand will be fulfilled

But for some region in e-government in which signals and internet connection are hardly found, it's better to apply simple help desk, generally used and know, and doesn't need internet connection, such as call center and text messaging. To implement such help desk like OA that need a strong connection is not suggested, because it can reduce service effectiveness and spent more money than needed.

If it is difficult to maintain and manage help desk on its own, government can outsource help desk service with considerations before. By outsourcing, government deliver the task to some organization outside e-government, so it will double the task to train the employees before they can become operator for help desk service. But the good side is, by outsourcing the maintenance expense can be reduced and some kind of help desk like application will be more user-friendly because it's handled by the expert.

To find out whether the service quality is good and effective, e-government can use the criteria above to measure whether the help desk have good ease of use, whether it is trusted by citizens enough or not, whether it support the citizens or not, and by the interactivity of it.

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